Reward-Channel Separability in Multimodal Chart RLVR: A Cross-Rollout Bonus and a Negative-Sample Caveat

Anonymous ACL submission

Abstract

Chart question answering requires a model to parse the visual structure, extract the relevant data, and reason to a final answer. Whether these components respond independently to different training signals, however, is not well understood. We study this with two opposing interventions: Hierarchical Correct-Path Consistency (**HCPC**), a cross-rollout bonus rewarding consistent extraction and diverse reasoning among correct responses, and Negative Sample Reinforcement (**NSR**; **?**), which removes gradient signal from correct responses entirely. We evaluate on ChartQA (in-distribution) and the held-out ChartFC benchmark, which is excluded from training and serves as our out-of-distribution (OOD) test. NSR collapses format compliance from 96.8% to 31.6% on ChartQA with no gain in answer accuracy over GRPO. Per-tag analysis traces the failure to the structural tags reward training had to teach, pointing to reward-signal starvation as the cause. HCPC achieves the best Pass@4 on ChartQA (0.828 *vs.* 0.816 for GRPO) but degrades on ChartFC (0.918 *vs.* 0.922 for GRPO), where reliance on structured extraction becomes a liability. Combining the two, NSR+HCPC recovers GRPO-level accuracy on ChartFC while format compliance barely recovers. The fact that accuracy returns but format does not suggests the two are distinct behavioral channels, and that reward update rules from single-component settings should be re-evaluated before applying them to richer reward landscapes. We will make our code and data public.

1 Introduction

Recent chart-reasoning vision-language models are trained via reinforcement learning with verifiable rewards (RLVR), typically using GRPO with multi-component rewards. Answering questions about charts requires two sequential operations that are rarely treated as distinct (**????**). A model must first extract the underlying data table from the visual layout, then reason over that table to reach an answer. The first operation has one correct outcome; the second admits many valid paths. Current training pipelines collapse this distinction. The dominant recipe applies GRPO (**?**) with verifiable rewards across chart-type identification, table reconstruction, process conformity, and answer accuracy (**????**), reinforcing all correct rollouts uniformly regardless of which operation they got right or how. The practical cost is gradual: the policy over-commits to whichever solution paths dominate the training distribution, eroding the diversity that supports robustness under distribution shift and coverage at higher K in Pass@K (**?**).

We propose Hierarchical Correct-Path Consistency (**HCPC**), a cross-rollout bonus that treats the two operations differently. Among rollouts that get both the table and the answer right, HCPC rewards consistent table extraction and diverse reasoning traces. Consistency is appropriate at the extraction layer, where there is one correct table and reliable recovery is the goal. Diversity is appropriate at the reasoning layer, where multiple paths to the same answer reflect genuine competence. A ground-truth-anchored filter ensures the bonus targets correct-path quality rather than surface agreement among arbitrary rollouts.

To explicitly isolate what HCPC is and is not doing, we introduce a contrastive diagnostic that applies opposing pressure to the same problem. Negative Sample Reinforcement (NSR; **?**) follows an opposite optimization strategy by masking gradient signal on correct rollouts while penalizing incorrect ones. NSR was originally motivated by mitigating reward hacking and overconfident convergence on easy solutions, though its mechanism may also encourage greater exploration and diversity. In single-component math RLVR, this strategy is effective (**?**). In chart RLVR, however, we do not observe comparable gains. Port-

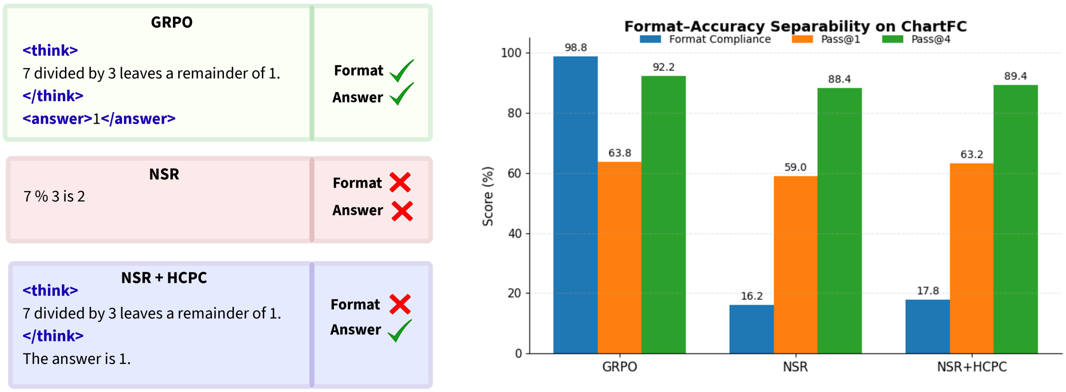

Figure 1: Format and answer accuracy are separable channels in chart RLVR. **Left:** a stylized rollout under each training regime. GRPO emits the required structural tags and the correct answer; NSR drops both; NSR+HCPC recovers the answer while the structured tags remain collapsed. **Right:** ChartFC (OOD) metrics for the same three regimes. Adding HCPC on top of NSR restores Pass@1 and Pass@4 to near GRPO levels, but format compliance barely moves from NSR's collapsed value (16.2% to 17.8%). Accuracy returns; format does not. The dissociation is direct evidence that these two behaviors respond independently to training interventions.

ing NSR unchanged collapses format compliance, specifically the structured XML tags and table markup that the pipeline depends on, on both in-distribution and out-of-distribution data, with no corresponding gain in answer accuracy. Per-tag analysis traces the failure to exactly the structural tags that reward training had to teach, pointing to reward-signal starvation as the cause. The result is informative precisely because the failure is clean: something about the multimodal, multi-component reward landscape breaks an assumption that held in the simpler setting. Adding HCPC to NSR recovers competitive answer accuracy out-of-distribution while format compliance stays near zero. Accuracy returns; format does not. This dissociation is the central finding of the paper: format compliance and answer accuracy are distinct behavioral channels in chart RLVR, and they can be manipulated independently. The implication is that reward channels in this setting are separable and should be designed with that separability in mind. Our contributions are as follows:

1. **HCPC (§3.2, §5.2).** A cross-rollout bonus applying level-specific pressure to table extraction and reasoning separately. HCPC improves rollout-set robustness (Pass@4, all-rollouts-correct rate, format compliance) while maintaining parity with the GRPO baseline on single-sample accuracy.

2. **NSR transferability analysis (§5.3).** Porting NSR unchanged into our multi-component chart-reasoning training pipeline collapses format compliance without accuracy gain. Per-tag analysis identifies reward-signal starvation as the mechanism, localized to structural tags the base model does not emit unprompted.

3. **Separability of format and accuracy (§5.4).** Adding HCPC to NSR recovers out-of-distribution accuracy all the way back to the GRPO baseline, while format compliance barely moves from NSR's collapsed level. One channel snaps back; the other does not. This is direct evidence that the two are distinct behavioral channels that respond independently to training interventions. We argue reward components in chart RLVR should be designed with this separability explicit.

2 Related Work

Chart reasoning with RLVR. Most current chart-reasoning VLMs are trained with GRPO and multi-component verifiable rewards. Chart-RVR

(?) set the template, combining rewards for format, chart-type, table reconstruction, process conformity, and answer accuracy. Recent work extends this along orthogonal axes: reasoning-trace supervision and CoT distillation (??), richer training data (??), grounding reflection (?), and agentic pipelines that wrap base models rather than modifying training (???). Our work sits next to these. We keep Chart-RVR's reward backbone unchanged and ask a different question: how the choice of advantage rule and a cross-rollout reward interact with that backbone.

Advantage-rule variants in RLVR. GRPO (?) treats correct and incorrect rollouts symmetrically, normalizing rewards within a group. ? split this into positive- and negative-sample reinforcement (PSR and NSR), and show that NSR alone, masking gradients on correct rollouts and penalizing only incorrect ones, matches or beats GRPO on math benchmarks. Related work studies RLVR in adjacent settings (????). All assume a single scalar reward, the setting NSR was designed for. We are the first to port NSR into a multi-component, multimodal reward landscape, and to identify a clean failure mode: structural rewards saturate early, pushing well-formed but answer-wrong rollouts above NSR's correctness threshold, which masks them and starves the structural side of positive gradient.

Group-level signals and rollout-set diversity. Self-consistency (?) is the closest existing cross-rollout signal, but operates at inference, aggregating K samples by majority vote for calibration, not training. ETTRL (?) preserves diversity at test time via entropy control. HCPC differs in three ways: it acts at training time as a reward bonus; it is anchored to ground-truth correctness, not majority agreement, so consistently wrong groups get no bonus; and it applies different objectives at different levels, consistency for table extraction and diversity for reasoning. We use Pass@K (?) as our coverage diagnostic; HCPC's level-specific design targets the higher-K regime where symmetric advantage rules fall behind.

Format and accuracy as separable behaviors. The dissociation we find, where format compliance and answer accuracy move in opposite directions under the same intervention, has no direct precedent in chart RLVR we are aware of. The closest adjacent observation is that CoT prompting can degrade VLM accuracy in chart settings (?), suggesting structured reasoning and correctness already trade off through some channel. Our result is stronger than a tradeoff: NSR+HCPC recovers GRPO-level accuracy while format compliance stays at NSR's collapsed level. This makes reward-component independence a real design choice for VLM RLVR, not a side note.

3 Methodology

We build on Chart-RVR (?) as a reward backbone and introduce two interventions: HCPC (Hierarchical Correct-Path Consistency), a group-level bonus rewarding level-specific structure among correct rollouts, and NSR (?), a contrastive gradient-masking strategy that we port to chart RLVR as a diagnostic.

3.1 Preliminaries: Chart-RVR Reward Backbone

We adopt Chart-RVR's per-rollout reward structure, which supervises each output component independently. Rollouts are expected in the format <think><type/> <table/> reasoning</think> <answer/>. Four components are active: R_{fmt} (structural compliance with the required XML template), R_{acc} (answer correctness against y^*), R_{tbl} (table extraction quality assessed by partial structural alignment against T^*), and R_{type} (exact match of the predicted chart type against the ground-truth type c^*). Their sum gives the base reward:

$$R_{\text{base}}(o_i) = R_{\text{fmt}} + R_{\text{acc}} + R_{\text{tbl}} + R_{\text{type}}.$$

For answer evaluation, we intentionally decouple extraction of $\hat{y}_i$ from full-format validity. We read $\hat{y}_i$ from the first <answer>...</answer> span if one is present, even if other parts of the output violate the XML schema; if the answer span is missing or malformed, we set $\hat{y}_i{=}\varnothing$ and mark the rollout incorrect. Thus a rollout can be answer-correct but format-incorrect, which is precisely the operational basis of the format/accuracy dissociation analyzed later. By contrast, HCPC's correct-path filter in §3.2.1 uses the stricter requirement that the entire response be structurally parseable.

Chart-RVR also defines a process reward measuring similarity to the ground-truth reasoning chain; we disable it because it conflicts with our diversity objective.

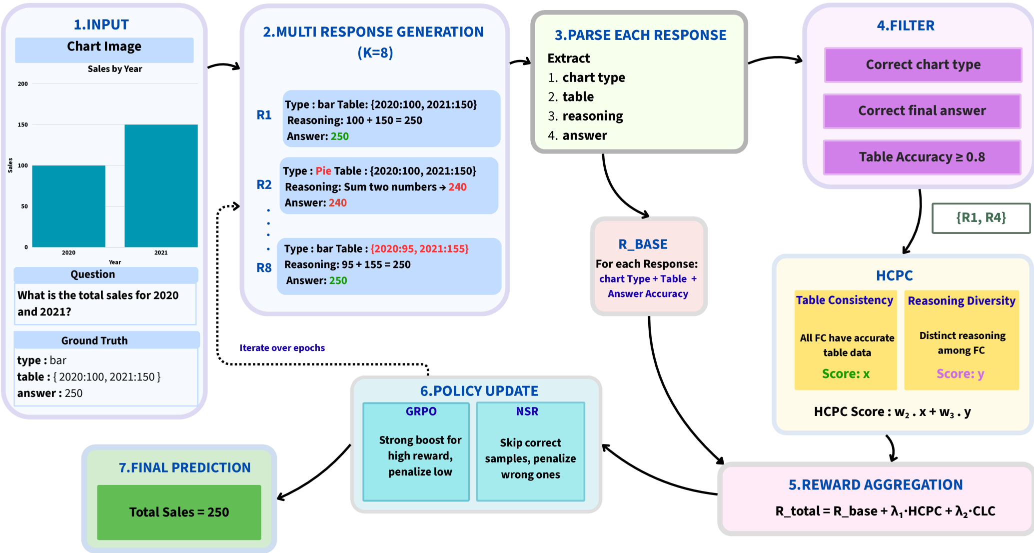

Figure 2: Training pipeline. The multi-component reward supervises each rollout independently through format, table, accuracy, and length rewards. HCPC adds a group-level bonus computed over rollouts that pass the correct-path filter (G^+), rewarding consistent table extraction together with diverse reasoning. We compare two advantage estimators over the same per-rollout reward: standard GRPO and NSR, which masks the gradient of correct rollouts.

3.2 HCPC: Hierarchical Correct-Path Consistency

HCPC augments R_{base} with a group-level bonus computed exclusively over correct rollouts. The design enforces *level-specific coherence*: consistency where one factually correct answer exists (table extraction) and diversity where multiple paths are valid (reasoning steps). Cross-rollout statistics are conditioned on ground-truth correctness rather than majority agreement (**?**), preventing shared extraction errors from producing a spuriously high consistency signal.

3.2.1 Correct-Path Filtering

Let $G = \{o_1, \ldots, o_K\}$ be the rollouts for a prompt with ground truth (T^*, y^*). We define the *correct-path subset* $G^+ \subseteq G$ via three sequential gates.

Gate 1 (format). The rollout must be structurally parseable with all required tags and a valid table block.

Gate 2 (table fidelity). The extracted table $\hat{T}_i$ must satisfy $\text{sim}(\hat{T}_i, T^*) \geq \delta$ with δ=0.8, using cosine similarity over MiniLM-L6 embeddings (**?**) on serialized table strings. This excludes rollouts that reach the correct answer from a misread table.

Gate 3 (answer). The predicted answer $\hat{y}_i$ must match y^*: relative error $\leq$5% for numeric answers, normalized substring match for string answers.

All three gates must pass. An answer-correct rollout with a wrong table has not followed a genuine reasoning path; rewarding its cross-rollout properties would propagate extraction errors into the group signal.

3.2.2 Level-Specific Group Metrics

On G^+ we compute three metrics targeting distinct trajectory levels. C_{type} is the fraction of rollouts in G^+ that predict the modal chart type $\hat{c}_{\text{mode}}$. For table and reasoning levels, let $\langle f \rangle_{G^+}$ denote the mean of $f(i, j)$ over all distinct pairs $i < j$ in G^+:

$$C_{\text{table}} = \left\langle \text{sim}(\hat{T}_i, \hat{T}_j) \right\rangle_{G^+},$$

$$D_{\text{reason}} = 1 - \left\langle \text{sim}(\hat{w}_i, \hat{w}_j) \right\rangle_{G^+}.$$

C_{table} measures pairwise agreement among extracted tables; correct rollouts should converge on the one ground-truth reading. D_{reason} is pairwise dissimilarity among reasoning strings $\hat{w}_i$; high values indicate diverse inferential paths to the same answer. C_{type} and C_{table} enforce convergence where the correct answer is unique; D_{reason} enforces divergence where multiple paths are valid. All similarities use cosine distance over MiniLM-L6 embeddings.

3.2.3 Reward Integration

The consistency component of the group bonus is shared equally among all rollouts in G^+:

$$B_{\text{cons}} = \frac{|G^+|}{K}\big(w_{\text{type}}\, C_{\text{type}} + w_{\text{table}}\, C_{\text{table}}\big).$$

The $|G^+|/K$ prefactor scales the consistency bonus by the correct-path fraction, so the signal grows with the share of rollouts the model already gets right and vanishes when the group is mostly incorrect. The diversity component is assigned per-rollout via a uniqueness score $u_i = 1 - \bar{s}_i$ for $i \in G^+$, where

$$\bar{s}_i = \frac{1}{|G^+| - 1} \sum_{j \in G^+,\, j \neq i} \text{sim}(\hat{w}_i, \hat{w}_j)$$

is rollout i's mean cosine similarity to the other reasonings in G^+. The rollout average of u_i over G^+ equals D_{reason}, so u_i is the per-rollout decomposition of the group-level diversity. For groups with $|G^+| \geq 2$, the per-rollout HCPC reward is:

$$R_{\text{HCPC}}(o_i) = \begin{cases} B_{\text{cons}} + w_{\text{reason}}\, u_i & \text{if } o_i \in G^+, \\ 0 & \text{otherwise,} \end{cases}$$

with w_{type}=1.0, w_{table}=2.0, and w_{reason}=1.5. The total reward is:

$$R_i = R_{\text{base}}(o_i) + R_{\text{HCPC}}(o_i).$$

Choice of weights. The weights are fixed rather than tuned, and the chosen values reflect the empirical magnitudes of each term so that no single component dominates trivially. C_{table} ranges in $[0, 1]$ but is the signal most directly aligned with correctness, so it receives the largest weight. C_{type} also ranges in $[0, 1]$ but is typically near ceiling on groups where HCPC fires (chart-type prediction saturates early in training), so it receives a smaller weight to avoid contributing a near-constant offset. The per-rollout uniqueness u_i has a much smaller empirical range ($\bar{u} \approx 0.05$ at K=4): w_{reason} is raised above w_{type} so that the diversity term remains a non-trivial part of the bonus despite its smaller magnitude. Even so, $w_{\text{table}}\, C_{\text{table}}$ remains the dominant contribution in practice.

We do not perform a weight sweep or per-component ablation: with three weights, K=4 rollouts, and full-pipeline training runs for every grid point, a proper isolation is beyond our compute budget. We acknowledge this as a limitation and leave systematic ablation of $(w_{\text{type}}, w_{\text{table}}, w_{\text{reason}})$ to future work. HCPC is silent when $|G^+|<2$; on ChartQA this occurs 28.8% of the time ($|G^+|$=0 on 17.2% of groups, $|G^+| \geq 3$ on 56.6%), exceeding the naïve Bernoulli bound of 15.7% at p=0.62 due to within-group visual correlation.

3.3 Diagnostic Baseline: NSR

Negative Sample Reinforcement (?) pursues the same diversity goal as HCPC by the opposite route: it removes gradient from correct rollouts entirely and penalizes only incorrect ones. In single-component math RLVR this improves Pass@K (?). We port it to chart RLVR unchanged to test whether that finding transfers to a multi-component reward setting.

Masking rule. After GRPO normalizes advantages within a group, we zero out the advantage of any rollout with $R_i \geq \theta$, where $\theta = 0.5\, R_{\max}$. $R_{\max}$ is summed from all defined reward ceilings in the Chart-RVR specification regardless of whether each component is active: $R_{\max}$=11.0 for the base configuration and $R_{\max}$=15.5 with HCPC (11.0 + w_{type} + w_{table} + w_{reason}), giving θ=5.5 and θ=7.75 respectively. Because two base components are inactive (§3.1), the achievable ceiling is lower; the threshold is therefore conservative, masking only the highest-scoring rollouts. The threshold is fixed rather than group-relative, so it does not shift with the per-step distribution.

NSR+HCPC. When combined, HCPC bonus inflates R_i for rollouts in G^+, making them more likely to be masked by NSR. Rollouts outside G^+ receive no HCPC bonus and are more likely to retain a negative gradient. HCPC membership thus indirectly determines which rollouts drive the policy update under NSR.

4 Experimental Setup

4.1 Data and Training Setup

Training uses the first 1,000 samples of `sanchit97/chart-rvr-grpo-train` (?), Chart-RVR's CoT training mixture drawn from ChartQA (?), PlotQA, and ChartFC (?). Samples are selected sequentially by index. ChartFC samples in the training mixture come from the training split of ?; the OOD evaluation probe (§4.2) uses the disjoint test split. PlotQA samples are included in training for domain coverage but excluded from evaluation, which targets chart QA and fact-checking directly.

We fine-tune Qwen2.5-VL-3B-Instruct with LoRA (r=8, α=16 on W_q, W_v) using AdamW at learning rate 1×10^{-6}, effective batch size 4 (per-device 1, gradient accumulation 4), K=4 rollouts per prompt at temperature 0.8 and top-p=1.0, over 2 epochs. No KL penalty is applied (β_{KL}=0). In our setup, the verifiable rewards themselves act as the main constraint against degenerate outputs, and LoRA (restricted to W_q, W_v with r=8) sharply limits how far the policy can drift from the base model, so an additional reference-policy KL term was empirically unnecessary on our 1,000-sample, 2-epoch budget. We evaluate five configurations: BASE (untuned Qwen2.5-VL-3B-Instruct), GRPO, GRPO+HCPC, NSR, and NSR+HCPC. GRPO uses the standard group-normalized advantage $\hat{A}_i = (R_i - \bar{R})/\max(1, s_R)$; NSR additionally zeros out advantages on rollouts with $R_i \geq \theta$ (§3.3).

4.2 Evaluation Protocol

Evaluation samples 4 generations per test prompt at temperature 0.8 and top-p=0.95, the standard inference-time setting; training uses top-p=1.0 to keep the rollout distribution unrestricted during exploration. We evaluate on two disjoint 500-sample slices, one per benchmark, drawn from the official test split of each. **ChartQA** (**?**) provides the in-distribution slice; **ChartFC** (**?**) provides the held-out OOD slice, differing from ChartQA on task format (fact-checking vs. open-ended QA) and visual style. The two slices are reported separately and never mixed.

We report Pass@1, unbiased Pass@4 (**?**), and format compliance Fmt%. Two additional metrics are central to this paper.

Per-tag emission rate. For each required tag (⟨think⟩, ⟨answer⟩, ⟨type⟩, ⟨table⟩, plus proper order), the fraction of rollouts emitting it. This localizes format collapse to specific structural components rather than reporting it in aggregate.

Table-dependence Δ_{tbl}.

$$\Delta_{\text{tbl}} = P(\hat{y} = y^* \mid \text{tbl parsed}) - P(\hat{y} = y^* \mid \text{no tbl}),$$

computed using *within-sample averaging*: for each test sample s we form the per-sample conditional means $\bar{p}_s^+ = \text{mean}_i\{\hat{y}_i = y^* : \text{tbl}_i \text{ parsed}\}$ and $\bar{p}_s^- = \text{mean}_i\{\hat{y}_i = y^* : \text{tbl}_i \text{ not parsed}\}$ over the K=4 rollouts of that sample. The slice-level Δ_{tbl} then averages each per-sample mean across the samples that have at least one rollout in the relevant condition. This keeps each test sample as the unit of observation (so paired-bootstrap CIs remain valid) while using all K rollouts to reduce per-sample noise, rather than relying on a single rollout index. A higher Δ_{tbl} indicates the model's answer correctness depends more strongly on emitting a parseable table. The two arms of the conditional can move for different reasons; we unpack this in §5.2.

5 Results and Analysis

5.1 Main Results

Table 1 reports the five configurations on ChartQA and ChartFC. GRPO training substantially improves both answer accuracy and format compliance over the untuned Base model: Pass@1 rises from 0.518 to 0.630 on ChartQA and Fmt% from 29.8 to 96.8. GRPO+HCPC matches GRPO on Pass@1 (0.622 vs. 0.630) and pulls ahead at higher K (Pass@4 0.828 vs. 0.816), consistent with HCPC's design goal of improving rollout-set coverage rather than single-shot accuracy. GRPO retains the strongest table-dependence on both benchmarks (Δ_{tbl}=+0.265 on ChartQA, +0.160 on ChartFC), indicating that the un-augmented advantage rule already routes answer correctness through the extracted table; HCPC trades a small amount of this dependence for the rollout-set robustness gain. NSR collapses format compliance to 31.6% on ChartQA and 16.2% on ChartFC with no accuracy gain; the per-tag analysis in §5.3 traces this to reward-signal starvation on structural tags. NSR+HCPC recovers OOD accuracy to 0.632 (statistically tied with GRPO at 0.638; §5.4) while format compliance stays at 17.8%, the format/accuracy dissociation that is the central finding of this paper.

5.2 GRPO+HCPC: Rollout Robustness Without Accuracy Regression

HCPC is designed to strengthen correct-path quality by rewarding consistent extraction and diverse reasoning. The metrics that should reflect this are rollout-set robustness and Pass@K at higher K, not single-shot Pass@1.

Headline Pass@1 on ChartQA is statistically tied: GRPO 0.630 vs. GRPO+HCPC 0.622 (paired bootstrap −0.8 pp, CI [−3.8, +5.4],

	ChartQA (in-distribution, n=500)				ChartFC (OOD)			
Method	Pass@1	Pass@4	Δ_{tbl}	Fmt%	Pass@1	Pass@4	Δ_{tbl}	Fmt%
Base	0.518	0.774	+0.005	29.8	–	–	–	–
GRPO	**0.630**	0.816	**+0.265**	96.8	**0.638**	**0.922**	**+0.160**	**98.8**
GRPO+HCPC	0.622	**0.828**	+0.144	**98.2**	0.606	0.918	+0.020	66.4
NSR	0.592	0.808	+0.041	31.6	0.590	0.884	−0.064	16.2
NSR+HCPC	0.574	0.800	+0.118	43.0	0.632	0.894	−0.003	17.8

Table 1: Main results. Δ_{tbl} is the conditional gain in answer correctness from emitting a parseable table; higher means stronger table-dependence. Bold marks column-best per benchmark. Pass@4 uses the unbiased estimator over 4 generations per prompt.

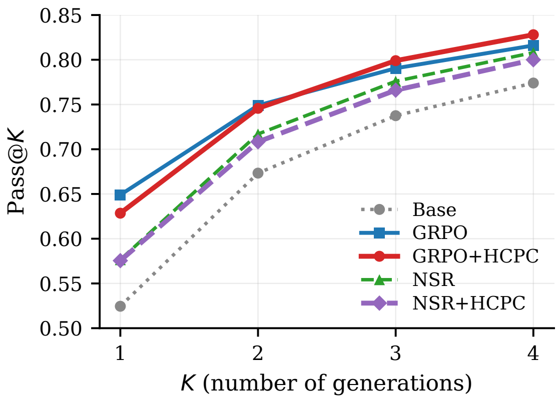

Figure 3: Pass@K on ChartQA (n=500). HCPC's gain over GRPO grows with K: tied at K=1, leading at K=3 and K=4. The direction is consistent with the design hypothesis that a cross-rollout bonus preserves correct-path coverage at higher K. Individual differences are not significant at n=500 (Pass@4 boot-p=0.48).

p=0.77, n_{boot}=10,000). On rollout-set robustness metrics, the direction favors HCPC: the fraction of test samples where all 4 rollouts are correct is 40.8% for HCPC vs. 38.6% for GRPO; the average correct rate per group is 0.628 vs. 0.620; Pass@4 is 0.828 vs. 0.816; format compliance is 98.2% vs. 96.8%. Table-dependence is the one direction where GRPO leads (Δ_{tbl}=+0.265 vs. HCPC's +0.144), which we unpack below. No individual difference reaches significance at n=500 (Pass@4 boot-p=0.48), but the joint direction across robustness metrics is consistent with the design hypothesis that a cross-rollout bonus preserves correct-path coverage without trading off single-shot accuracy. Figure 3 shows the Pass@K curves: HCPC matches GRPO at K=1 and pulls ahead at $K\geq$3, the regime where rollout-set coverage matters.

Unpacking Δ_{tbl}: $P(\hat{y} = y^* \mid \text{tbl})$ is 0.625 for GRPO and 0.634 for HCPC (tied on the common-table intersection, Appendix C), while $P(\hat{y} = y^* \mid \text{no tbl})$ is 0.360 for GRPO and 0.490 for HCPC. The two methods are indistinguishable when a parseable table is produced; the gap in Δ_{tbl} comes entirely from the no-table arm, where HCPC remains markedly more accurate than GRPO. The correct-path bonus appears to make the policy more robust to extraction failures rather than more dependent on the extracted table: when the table tag is missing, HCPC still answers correctly 49% of the time vs. GRPO's 36%.

GRPO+HCPC OOD behavior. On the OOD ChartFC probe, GRPO+HCPC trails plain GRPO on most metrics (Pass@1 0.606 vs. 0.638, Pass@4 0.918 vs. 0.922, format 66.4% vs. 98.8%). Format compliance is the largest drop: HCPC's extraction pipeline is less reliable on unfamiliar visual styles, and 34% of rollouts no longer emit a parseable table tag. Despite this, answer accuracy in the no-table arm holds up well ($P(\hat{y} = y^* \mid \text{no tbl})$=0.605 for HCPC vs. 0.500 for GRPO on the few GRPO samples that fall in this bucket), so the Pass@1 and Pass@4 gaps to GRPO remain small (−3.2 and −0.4 pp). Δ_{tbl} correspondingly drops to +0.020: with the no-table arm now containing a sizable share of the rollouts and the model still answering correctly without a table, the parseable-table condition no longer carries a large correctness premium. This is a genuine limitation of standalone GRPO+HCPC, complementary to the NSR-mitigation result in §5.4; we leave a fix (e.g., a chart-style augmentation in training or a softer GT-anchored filter) to future work.

5.3 NSR Breaks VLM Format

Switching the advantage estimator from GRPO to NSR, with no other changes, drops format compliance from 96.8% to 31.6% on ChartQA and from 98.8% to 16.2% on ChartFC. On ChartFC, GRPO leads NSR on Pass@4 by +3.8 pp (CI [+0.6, +7.2], p=0.028, paired bootstrap; McNe-

Method	think	answer	type	table	order
BASE	52.2	91.2	96.6	77.2	81.6
GRPO	97.4	97.6	100	98.6	97.4
GRPO+HCPC	98.8	98.2	100	99.6	98.6
NSR	57.4	94.8	96.6	72.2	85.4
NSR+HCPC	57.8	91.6	94.6	83.2	84.0

Table 2: Per-tag emission rate (%) on ChartQA. Column headers refer to the `<think>`, `<answer>`, `<type>`, `<table>` tags and the "proper order" check. NSR's collapse is concentrated in the structural tags (think, table) that the base model does not emit unprompted; HCPC under NSR partially recovers table but not think.

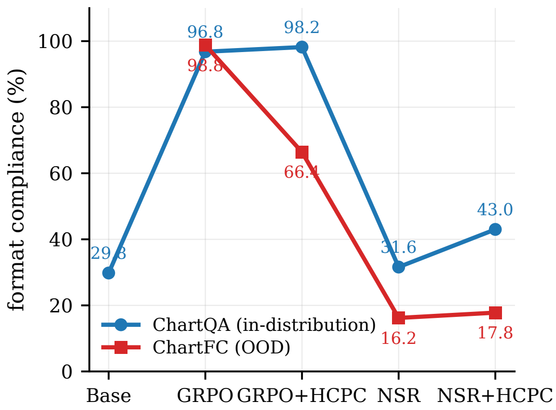

Figure 4: Overall format compliance per method on ChartQA (in-distribution) and ChartFC (OOD). NSR collapses on both; GRPO+HCPC holds ID but degrades OOD; GRPO is the only configuration that keeps format near ceiling under distribution shift.

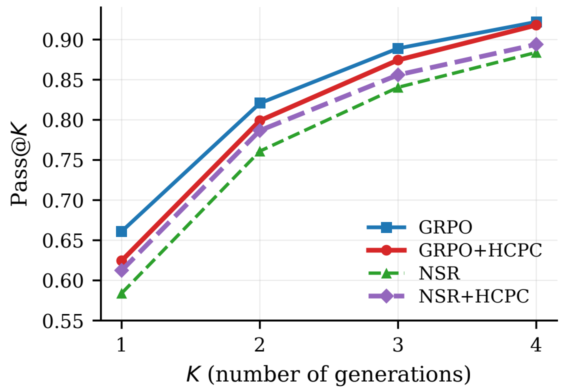

Figure 5: Pass@K on ChartFC (OOD, n=500). Plain NSR (green) sits below all other configurations at every K; adding HCPC (purple) closes most of the gap to GRPO. BASE was not evaluated on ChartFC.

mar exact p=0.034).

Table 2 localizes the collapse. NSR keeps the tags the base model already emits (⟨type⟩, ⟨answer⟩, proper order) and loses the ones GRPO has to teach (⟨think⟩, ⟨table⟩). NSR's ⟨think⟩ rate (57.4%) is essentially the base model's (52.2%) plus noise; ⟨table⟩ rate (72.2%) falls below the base model's (77.2%).

The mechanism is mechanical. R_{fmt}, the active structural reward in our setup, saturates quickly: emitting the required tags in the correct order is a learned-once behavior that yields near-maximal contribution from that component. With those rewards saturating, the total per-rollout R_i for a well-formed rollout reaches the NSR threshold $\theta = 0.5\,R_{\max}$ regardless of whether the final answer is correct. NSR therefore masks the advantage of *structure-good, answer-wrong* rollouts, leaving only *structure-bad, answer-wrong* rollouts, which receive negative advantage, to drive the policy. This removes the dominant positive gradient on structural-tag emission, and the policy drifts back toward the base model's unstructured output. We expect this failure mode to be absent in **?**'s math experiments because text-math rewards are dominated by a single answer bit; the implicit "format" (a final `\boxed{}` or a number on its own line) is either trivial or emerges from the answer-correctness signal itself. Verifying this empirically is beyond our current scope.

5.4 Format and Accuracy as Separable Channels

On ChartFC, HCPC added to NSR improves over plain NSR on every metric: Pass@1 $0.590 \to 0.632$ (+4.2 pp), Pass@4 $0.884 \to 0.894$ (+1.0 pp), Δ_{tbl} $-0.064 \to -0.003$ (+0.061), and format $16.2\% \to 17.8\%$. The Δ_{tbl} shift from clearly negative to near zero indicates that table extraction stops actively hurting; the format gain is marginal. Plain NSR is the weakest OOD configuration in Table 1 on every metric, and HCPC closes most of that gap (Figure 5). How the gap is closed (mechanism below) turns out to be more interesting than the gap-closing itself.

Format/accuracy dissociation. The way in which HCPC mitigates NSR's damage is unusual: NSR+HCPC reaches Pass@1 $=$ 0.632, statistically indistinguishable from GRPO (0.638; paired-bootstrap CI $[-0.046, +0.060]$, p=0.85), while format compliance remains collapsed at 17.8% (vs. GRPO's 98.8%). HCPC under NSR recovers GRPO-level OOD answer accuracy without recovering format. This is the cleanest empirical evidence in the paper that format compliance and answer accuracy are separable downstream behaviors in chart RLVR.

Why HCPC behaves so differently under GRPO and NSR. Under GRPO, HCPC behaves as designed: the bonus is added to correct-path rollouts' rewards, those rollouts get higher GRPO advantage $(R_i - \bar{R})/s_R$, and the policy is reinforced toward the correct path. This is the standard positive-gradient interpretation and accounts for the ChartQA results (§5.2).

Under NSR, the same bonus operates through a different channel. HCPC adds its bonus to exactly the correct-path rollouts that NSR then masks out of the gradient, so the bonus does not reach the policy as a direct positive update. But TRL computes the GRPO advantage before the NSR mask is applied, using the group mean $\bar{R}$ over the full set of rollouts. Adding R_{HCPC} to correct rollouts raises $\bar{R}$, and for the wrong rollouts that survive the mask this makes their advantage more negative. The correct-path bonus therefore reshapes the reference baseline that the surviving (wrong) rollouts are penalized against, pushing the policy harder away from wrong-path behavior. The mechanism is *baseline shift, not direct reinforcement*: HCPC under NSR strengthens the negative gradient on wrong rollouts without providing any positive gradient on correct ones. This is consistent with what we observe. Answer-level OOD accuracy recovers because wrong-rollout suppression strengthens, while structured-output emission, which requires positive gradient on correct rollouts that neither NSR nor HCPC-under-NSR can provide, remains collapsed.

On ChartQA, adding HCPC on top of NSR also partially restores format (31.6% $\rightarrow$ 43.0% overall, with the recovery concentrated in ⟨table⟩: 72.2% $\rightarrow$ 83.2%, Table 2). In-distribution, format partially recovers. Out-of-distribution, accuracy recovers fully while format does not. This asymmetry makes the channel-separability claim concrete.

6 Conclusion

We introduced HCPC, a cross-rollout bonus that rewards consistent extraction and diverse reasoning on the correct-path subset of a group. On top of GRPO it produces more robust correct groups and stronger table grounding without sacrificing Pass@1. Comparing against NSR, a diversity-preservation recipe imported from text-only math RLVR, we find that porting it unchanged into chart RLVR collapses format compliance: structural rewards saturate fast enough for well-formed rollouts to cross NSR's threshold regardless of answer quality. HCPC on top of NSR recovers GRPO-level OOD accuracy at near-zero format compliance, providing direct evidence that format and answer accuracy are separable channels of the trained policy. The takeaway for practitioners: advantage-rule asymmetries developed for single-reward source domains can interact destructively with reward components that are new in the target domain. The shape of the multi-component reward mixture deserves first-class consideration when porting RLVR recipes across modalities.

Limitations

We use a single 3B-parameter backbone (Qwen2.5-VL-3B), a 1,000-sample training subset, K=4 rollouts, and a 500-sample test slice per benchmark. Compute constraints prevented us from training on the full Chart-RVR CoT mixture (34,194 samples), evaluating on the full ChartQA and ChartFC test sets, or running EvoChart (**?**) and ChartQAPro (**?**) as style-OOD benchmarks distinct from the joint task-and-style probe ChartFC provides. Scaling to the full training data, full test sets, and a style-only OOD benchmark is our highest-priority follow-up.

A Reproducibility

Training and evaluation code is included with the submission. Hyperparameters and the exact reward configuration used for every run are specified in the training configs (`configs/experiment.py`); the expected output schema is in §B. Pass@K uses the unbiased estimator of **?**. Cross-rollout similarities use MiniLM-L6 embeddings. Paired-bootstrap CIs use n_{boot}=10,000 resamples; McNemar tests use the two-sided exact binomial form on the discordant pair.

B Prompt Template and Expected Output Format

The model is prompted to emit responses in a structured XML template. The prompt instructs the model to wrap its reasoning in <think>...</think>, identify the chart type, extract the underlying data table, perform step-by-step reasoning, and finally emit the answer in

<answer>...</answer>. The expected output schema is:

```
<think>
<type>...</type>
<table>{"columns": [...], "rows":
↪   [...]}</table>
<step-1>...</step-1>
<step-2>...</step-2>
...
</think>
<answer>...</answer>
```

The per-tag emission rate metric (§4.2) and the format reward R_{fmt} are both defined relative to this schema. <think> and <table> are not natively emitted by Qwen2.5-VL-3B-Instruct without training; the format reward provides the gradient that teaches them.

C Per-Tag Format Compliance, Both Benchmarks

Table 2 in the main body reports per-tag emission rates for ChartQA. Table 3 extends this to both benchmarks. BASE was not evaluated on ChartFC.

Method	think	answer	type	table	order
ChartQA (in-distribution)					
BASE	52.2	91.2	96.6	77.2	81.6
GRPO	97.4	97.6	100	98.6	97.4
GRPO+HCPC	98.8	98.2	100	99.6	98.6
NSR	57.4	94.8	96.6	72.2	85.4
NSR+HCPC	57.8	91.6	94.6	83.2	84.0
ChartFC (out-of-distribution)					
GRPO	99.0	99.0	100	99.6	99.0
GRPO+HCPC	78.0	92.4	96.6	87.2	79.0
NSR	51.4	91.0	96.2	55.6	56.0
NSR+HCPC	49.2	90.2	95.6	61.4	56.0

Table 3: Per-tag emission rate (%) on both benchmarks. On ChartFC, the <think> and <table> tags collapse hardest under NSR variants (51.4–61.4%, vs. GRPO's 99.0–99.6%). GRPO+HCPC also degrades OOD but milder (∼78–87%), in line with the OOD analysis in §5.2.

D Pairwise Statistical Comparisons

We computed paired-bootstrap 95% confidence intervals (n_{boot}=10,000) and exact two-sided McNemar p-values for all 20 method-pair × Pass@K × benchmark comparisons. Among the trained configurations (GRPO, GRPO+HCPC, NSR, NSR+HCPC), two pairwise gaps reach $p < 0.05$: GRPO vs. NSR on ChartFC Pass@4 (+3.8 pp, p=0.034; reported in §5.3) and GRPO vs. NSR+HCPC on ChartQA Pass@1 (+5.6 pp, p=0.029). All comparisons involving the untuned BASE model are significant on Pass@1 ($p < 0.05$). All other pairs, including the headline GRPO vs. GRPO+HCPC comparison (−0.8 pp on Pass@1, p=0.80; +1.2 pp on Pass@4, p=0.52), do not reach significance at α=0.05.

E HCPC Firing-Rate Distribution

HCPC's group bonus is zero when fewer than two rollouts pass the correct-path filter ($|G^+| < 2$). The empirical distribution of $|G^+|$ across the 500 ChartQA test groups for the trained GRPO+HCPC model is:

$\lvert G^+\rvert$	0	1	2	3	4
fraction	17.2%	11.6%	14.6%	15.8%	40.8%

HCPC fires (i.e., contributes a non-zero bonus) on groups with $|G^+|\geq 2$, which is $1 - (17.2 + 11.6) = 71.2\%$ of groups. The remaining 28.8% of groups receive no HCPC signal.

To check whether this silent rate is just driven by per-rollout accuracy, we compare it to the rate predicted under independence. The per-rollout probability of a rollout entering G^+ is the mean of $|G^+|/K$ across groups, which is p=0.628. If rollout-level correctness were independent within a group, the silent rate $P(|G^+|<2)$ would be $(1-p)^4 + 4p(1-p)^3 = 15.7\%$, and the all-wrong rate $P(|G^+|=0)$ would be $(1-p)^4 = 2.1\%$. The observed values (28.8% silent, 17.2% all-wrong) are roughly 1.8× and 8× the independence bounds respectively, confirming strong positive correlation of correctness within a group: rollouts on the same chart image and question tend to succeed or fail together.

F Δ_{tbl} on the Common-Table Subset

To rule out the confound that Δ_{tbl} is computed on different denominators per method, we recompute the table-conditioned arm on the intersection of samples where both methods produce at least one parseable-table rollout. On the 498 ChartQA samples where both GRPO and GRPO+HCPC have a parseable table available, $P(\hat{y} = y^* \mid \text{tbl})$ is 0.623 for GRPO and 0.634 for HCPC: indistinguishable. The two methods reach the same answer accuracy when a parseable table is produced. The Δ_{tbl} gap in Table 1 therefore arises entirely from the no-table arm. On the full slice, $P(\hat{y} = y^* \mid \text{no tbl})$ is 0.360 for GRPO and 0.490 for HCPC, so HCPC is more robust to extraction failures rather than more dependent on the extracted table for correctness.

G Per-Answer-Type Breakdown

We split ChartQA test samples by ground-truth answer type and compare GRPO against GRPO+HCPC on Pass@1. Differences are within exact-McNemar noise on numeric (n=302, -1.7 pp, p=0.63), boolean (n=75, $+4.0$ pp, p=0.71), and string (n=114, -6.1 pp, p=0.30). HCPC does not provide a category-specific accuracy advantage at this scale; the rollout-set robustness gains reported in §5.2 are uniform across answer types rather than concentrated on any single category.

H Qualitative Example: Format Collapse in Action

Illustration of the NSR collapse and HCPC partial recovery on a single ChartQA sample. The question is *"How many categories are there in the chart?"* with ground truth $y^* = 4$. We show rollout 0 from three methods.

GRPO (correct, well-formed).

```
<think><type>pie</type><table>{..., "rows":
↪  [["Asians","63"],...]}</table>
<step-1> The chart is a pie chart...
<step-4> Based on the information provided,
↪  there are four distinct categories...
</think>
<answer>4</answer>
```

NSR (format-collapsed; arrives at right reasoning but wrong answer surface form).

```
<think>First, I'll identify the type of chart
↪  and the underlying data table...
<step-1> The chart shows four categories:
↪  Asians, Whites, Blacks, and Hispanics. ...
Final answer:
Four
</think>
<answer>Four</answer>
```

NSR's rollout contains no `<table>` tag, no `<type>` tag, and emits the reasoning steps inline rather than as structured items. The answer is also expressed in word form ("Four") rather than the numeric form the eval expects ("4"), so the strict-match metric marks it incorrect despite the reasoning being substantively right.

NSR+HCPC (table tag recovered; answer still in word form).

```
<think><type>bar</type><table>{"columns":
↪  ["Country","Bar Color"], "rows":
↪  [["Asians","..."],...]}</table>
<step-1>From the chart...
<step-3>... there are four distinct
↪  categories.
</think>
<answer>Four</answer>
```

HCPC's group-level bonus on C_{table} pulls NSR's policy back toward emitting a `<table>` tag (table fraction $72.2\% \rightarrow 83.2\%$ on ChartQA, Table 2). The chart type is incorrectly identified (`bar` instead of `pie`), illustrating that HCPC's partial format recovery is not the same as a full quality recovery, and the answer is still "Four" in word form. This is exactly the dissociation pattern of §5.4: format is partly restored, but the answer-surface alignment that drives strict-match accuracy is governed by a separate channel.